

Air University



Computer Networks (Lab)

Year 2025

Mr. Waseem Younas

Academic Year: 2024 – 2028

Department of Cyber Security

USAMA ARSHAD | 247141

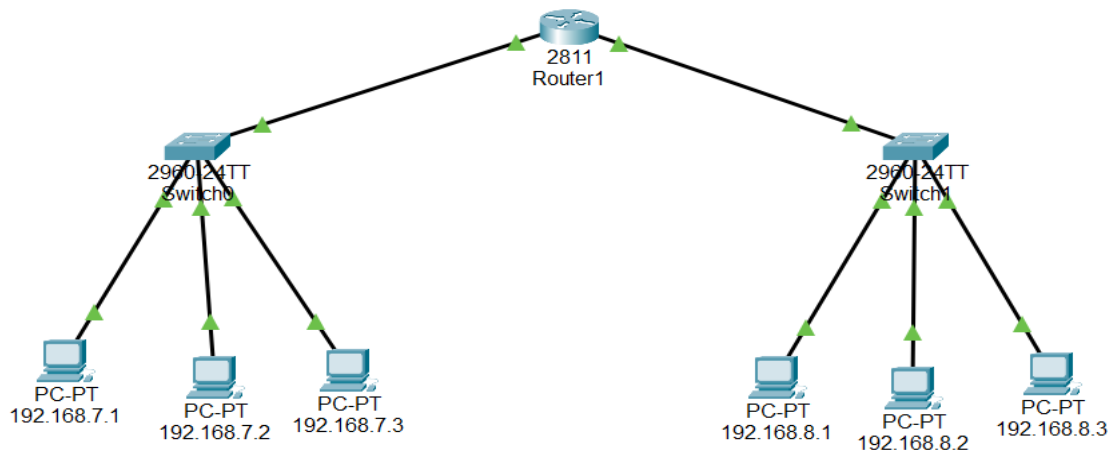
BS – Cyber Security – Fall – 24 (A)

Date of Submission: October 16, 2025

Lab Task # 04

Network Diagram:

This diagram illustrates a simple wide area network connecting two distinct local area networks via a central router. Each LAN operates on a different network ID, requiring a router for communication between the two segments.



Configuration:

The configuration lists the Layer 3 (IP) and Layer 2 (MAC) addresses for the critical devices and interfaces involved in routing a packet from one LAN to the other. The router acts as the gateway for both subnets.

A. Router

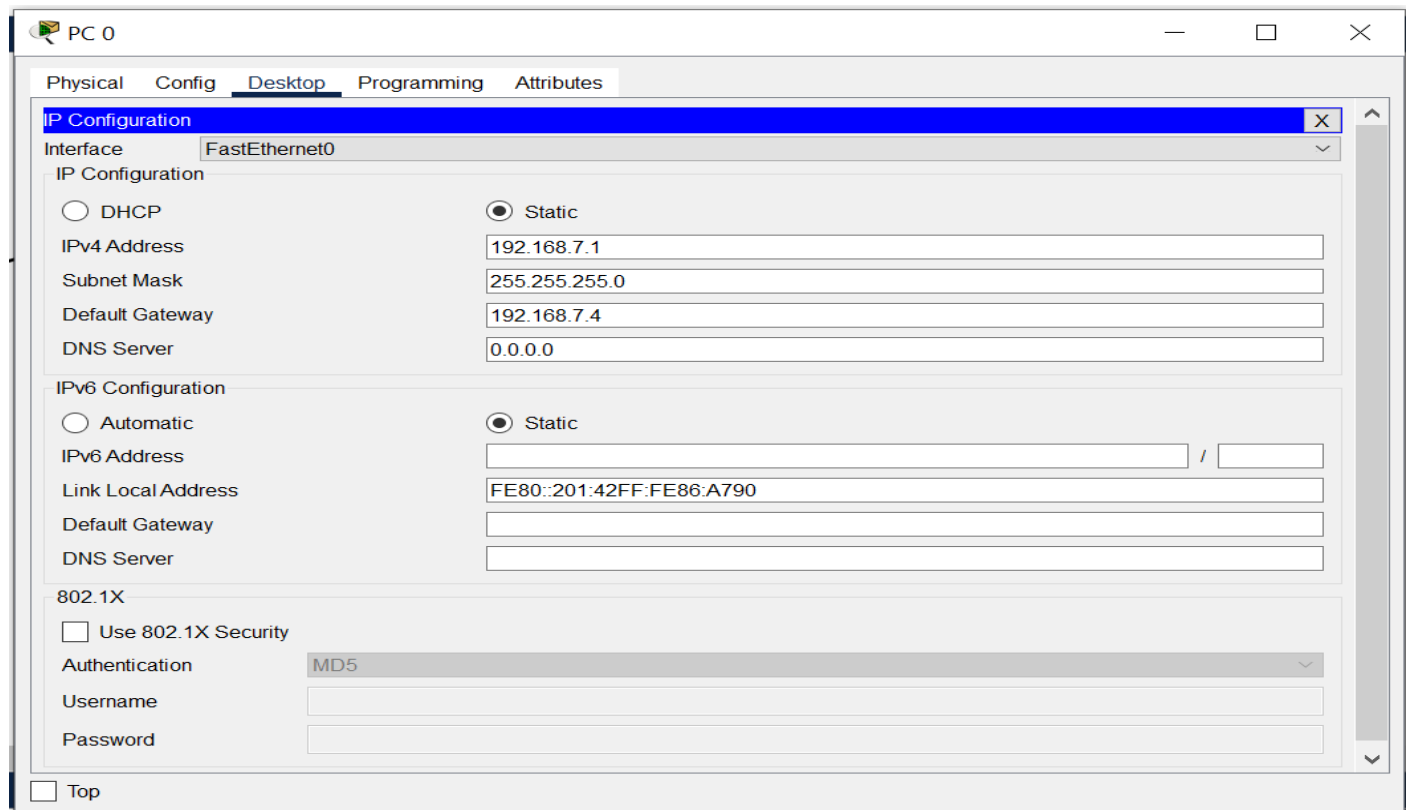
The screenshot shows the configuration window for Router1. The 'Config' tab is selected, and the 'FastEthernet0/0' interface is configured. The configuration includes the following fields:

- Port Status: On (checked)
- Bandwidth: 100 Mbps (selected)
- Duplex: Full Duplex (selected)
- MAC Address: 0050.0FEB.0601
- IP Configuration:
 - IPv4 Address: 192.168.7.4
 - Subnet Mask: 255.255.255.0
- Tx Ring Limit: 10

Below the configuration fields, the 'Equivalent IOS Commands' are listed:

```
Router1
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

B. Node



PC 0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.7.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.7.4

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:42FF:FE86:A790

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

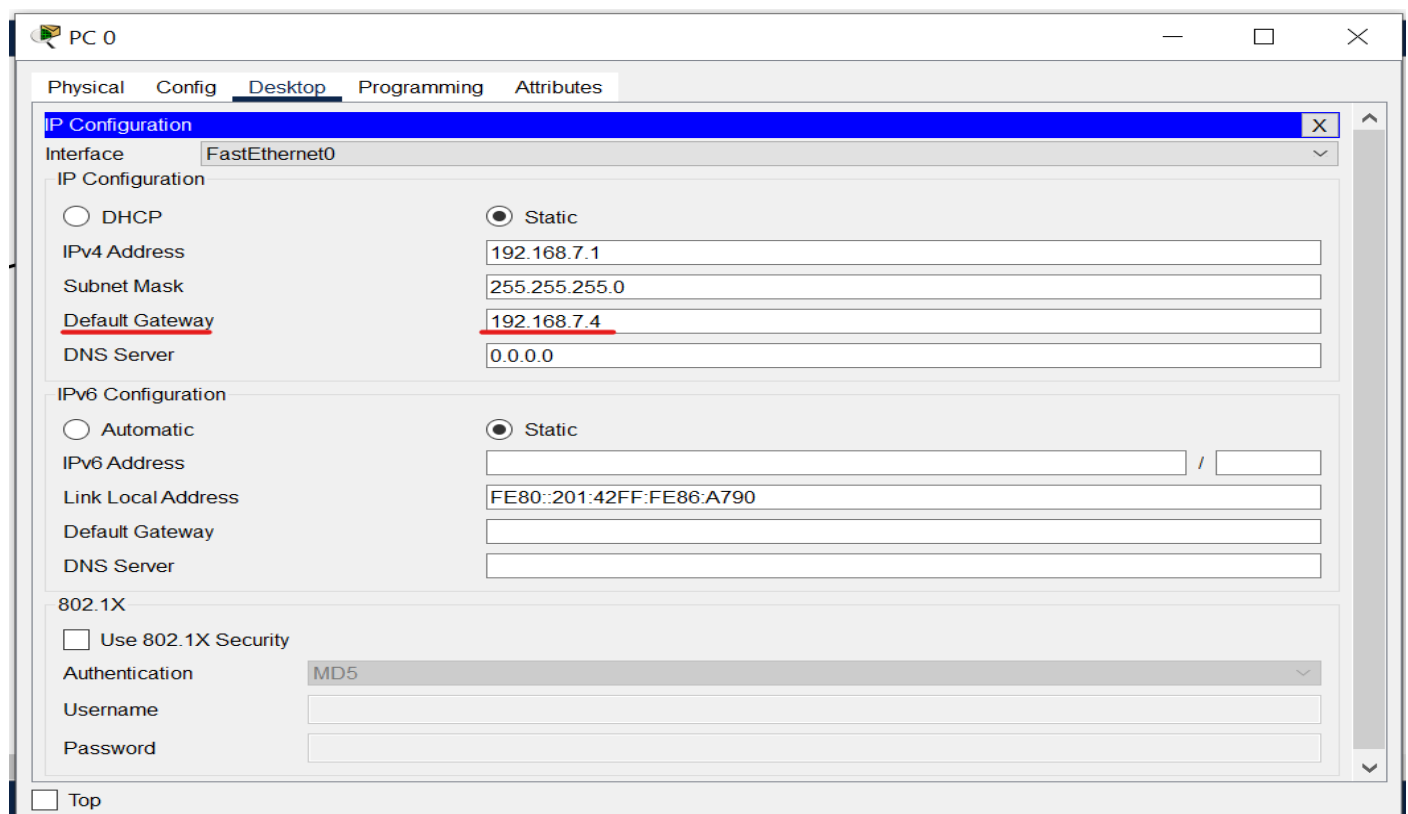
Username:

Password:

☐ Top

Gateway of Nodes:

The gateway is the IP address of the router port that connects the local network segment to the wider internet or other networks. It is the exit point for any packet destined for a different network ID.



PC 0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.7.1

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.7.4

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::201:42FF:FE86:A790

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

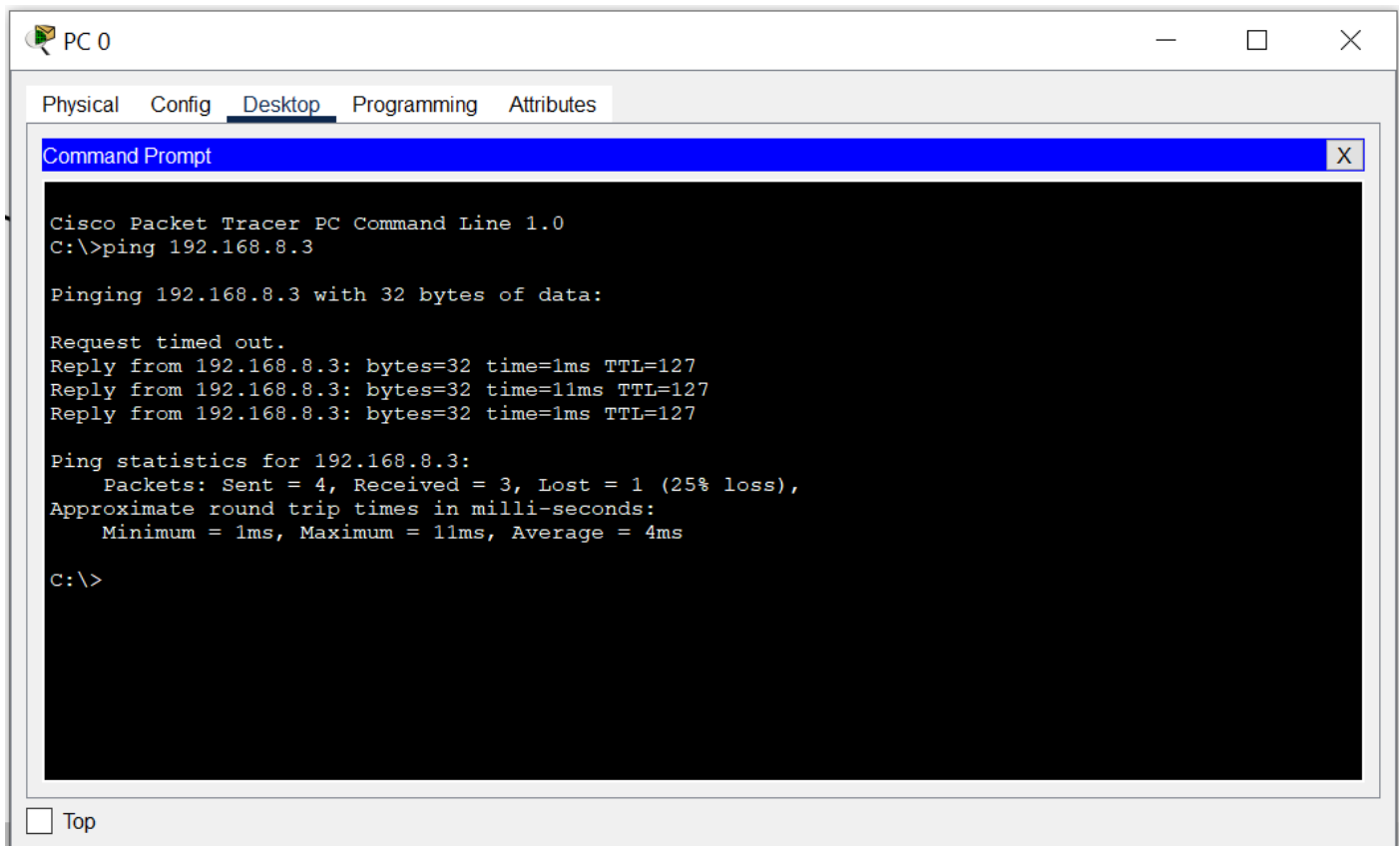
Username:

Password:

☐ Top

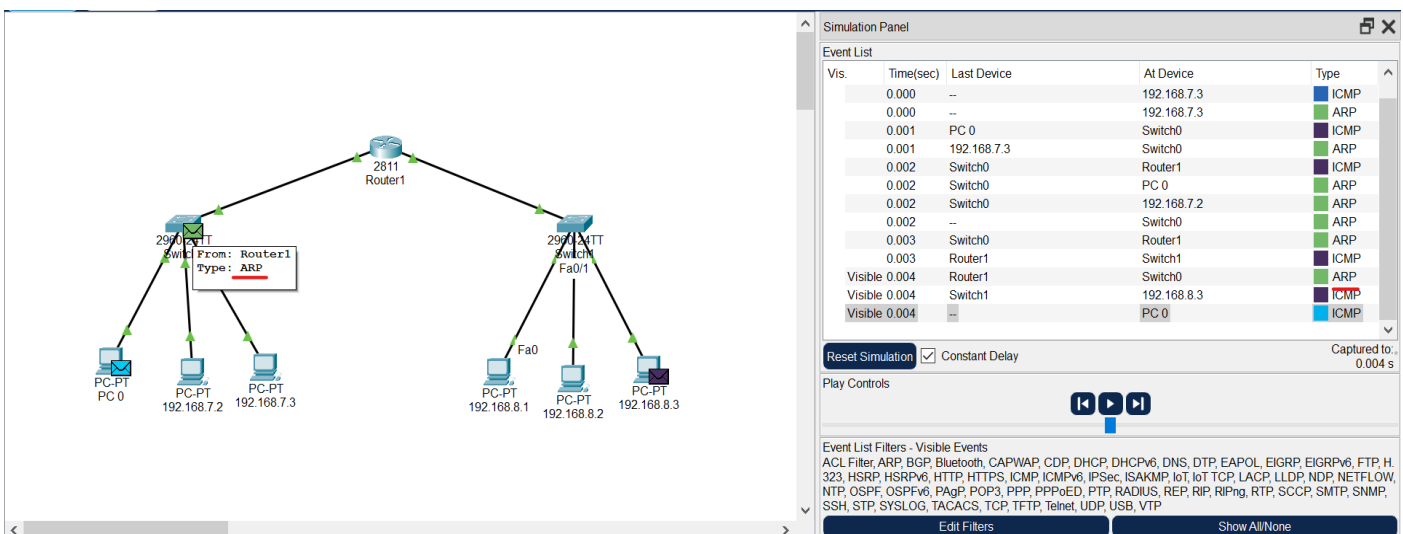
Ping:

A ping from PC 192.168.7.1 to 192.168.8.3 represents the initial packet (usually an ICMP Echo Request) sent across the networks.



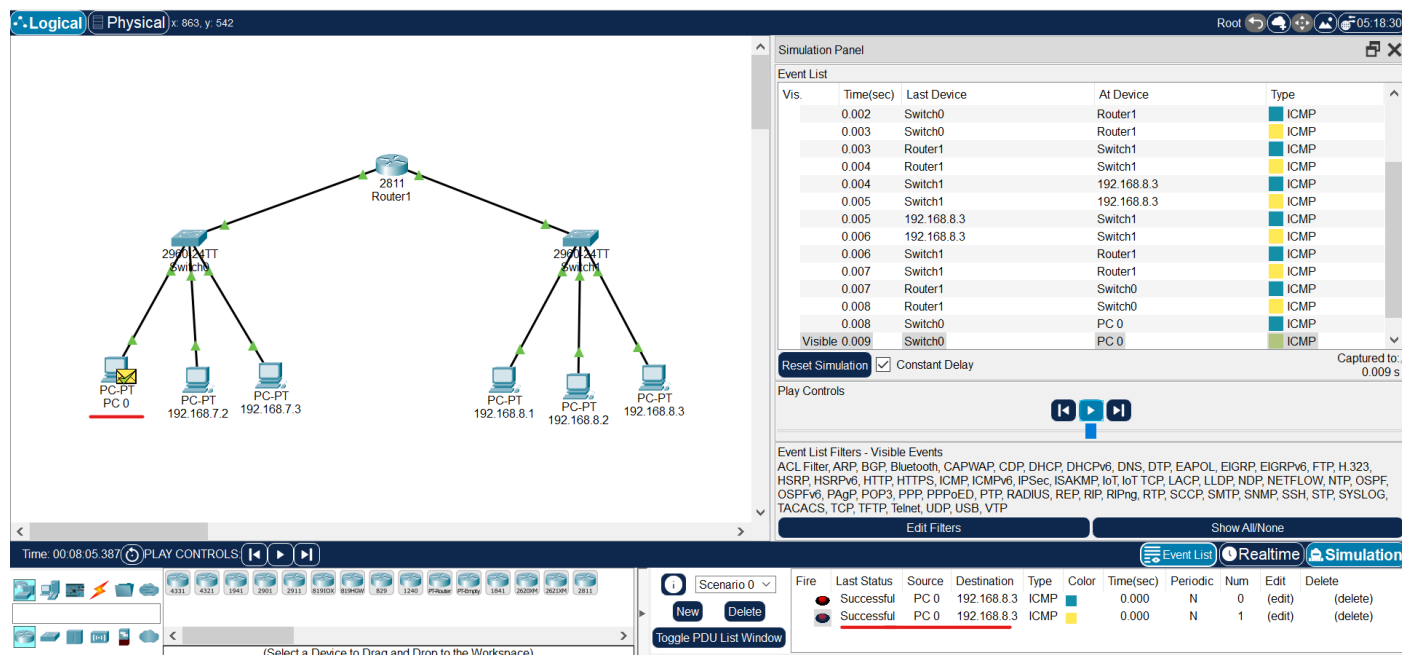
ARP:

Since the destination (192.168.8.3) is on a different network, the source PC sends its packet to the gateway (192.168.7.4). ARP is used twice to map the necessary IP addresses to MAC addresses.



Successful Transmission:

A successful transmission is achieved when the packet reaches the destination PC (192.168.8.3) and the subsequent reply successfully returns to the source PC (192.168.7.1).



THE END